Green Roofs for Flood Mitigation: The Case of Kiseana Neighborhood, Queens Borough

Data Overview

Project Description

Urban flooding has become one of the most pressing environmental challenges facing cities in the northeastern United States, and New York City's Queens Borough is no exception. The Kisena neighborhood, situated within a low-lying topographic zone and surrounded by impervious urban surfaces, experiences recurrent stormwater flooding events that overwhelm the existing gray infrastructure. Climate projections indicate that the frequency and intensity of extreme precipitation events will continue to rise, making proactive green infrastructure interventions both urgent and necessary.

This project conducts a GIS-based suitability analysis to identify rooftops in the Kisena Park area of Queens, New York, that are most suitable for green roof installation as a nature-based solution to urban flooding. Green roofs — vegetated layers installed on building rooftops — absorb and retain rainwater, reducing the volume and velocity of stormwater runoff that enters the drainage system. By strategically targeting buildings with the highest suitability for green roof retrofitting, this project aims to provide actionable spatial recommendations for urban planners, community stakeholders, and environmental policymakers seeking to enhance stormwater management and climate resilience in flood-prone neighborhoods.

The suitability analysis incorporates multiple spatial criteria including building rooftop area, structural load capacity indicators, solar radiation exposure, proximity to water bodies and drainage infrastructure, current land use classification, and building age. Each criterion is weighted and combined through a multi-criteria evaluation (MCE) framework in ArcGIS Pro to produce a composite suitability score for each building footprint within the study area. The results are then visualized through a series of interactive Tableau dashboards that allow users to explore suitability patterns across the neighborhood at multiple spatial scales.

Need for the Data

A robust spatial suitability analysis requires diverse, high-quality datasets that collectively capture the physical, structural, hydrological, and environmental characteristics of the study area. Each dataset plays a specific and indispensable role in the analysis:

> **Building Footprints and Rooftop Characteristics:** Rooftop area and structural attributes are fundamental to determining whether a building can physically support a green roof system. Buildings with larger, flatter rooftops and sufficient load-bearing capacity represent the highest-priority candidates. Without detailed building footprint data, it is impossible to

identify and rank candidate structures or estimate the potential stormwater retention capacity of green roof installations across the study area.

Digital Elevation Model (DEM) and Topographic Data: Elevation data is critical for understanding the hydrological flow paths of stormwater runoff and identifying low-lying areas most susceptible to flooding. A high-resolution DEM allows the analysis to map flood-prone zones within the Kisena neighborhood and to prioritize green roof placement in upstream areas where runoff retention will have the greatest downstream impact on flood reduction.

Solar Radiation Data: Green roofs require adequate sunlight for vegetation growth and maintenance. Solar radiation data helps identify rooftops with sufficient sun exposure to sustain plant life year-round, ensuring that recommended installations will remain ecologically viable and continue to provide stormwater benefits over their operational lifespan.

Land Use and Zoning Data: Land use classification helps filter buildings by ownership type and intended use, enabling the analysis to focus on residential, commercial, or institutional structures where green roof implementation is most feasible from a planning and regulatory perspective. Zoning data also informs the legal and administrative context within which retrofit recommendations can be acted upon.

Stormwater Infrastructure and Drainage Networks: Data on existing storm sewer networks, combined sewer overflow (CSO) outfalls, and drainage catchment boundaries is essential for understanding where the stormwater system is most overburdened. Prioritizing green roofs near these critical infrastructure points maximizes the direct flood mitigation benefit of each installation and helps model the expected reduction in peak flow volumes.

Building Age and Permit Records: Older buildings may have structural limitations that reduce their suitability for green roof retrofitting. Building age and permit data provide a proxy for structural condition and help flag buildings that may require additional engineering assessment before installation, allowing the suitability model to account for feasibility constraints beyond simple physical dimensions.

Data Sources

All datasets used in this project are publicly available through governmental and institutional open data portals. The table below summarizes the key datasets, their sources, and their role in the analysis.

Table 1: Summary of datasets, their sources, and analytical role in the green roof suitability analysis.

Spatial Data	Type	Projection	Source	Statistics
DSM	Tif	NAD 1983 StatePlane New York Long Isl FIPS 3104 (US Feet)	USGS Earth Explorer	
Building Footprints	Vector	NAD 1983 StatePlane New York Long Isl FIPS 3104 (US Feet)	National Register Building Listing-Nyc.gis.ny.gov	Mean Area: 468.56 sq.m Min: 48.11 sq.m Max:27, 399.45
Water Availability	Vector	NAD 1983 StatePlane New York Long Isl FIPS 3104 (US Feet)	US Environmental Protection Agency	Mean: 5, 365, 020, 598.88 Sum:155, 662, 250.352

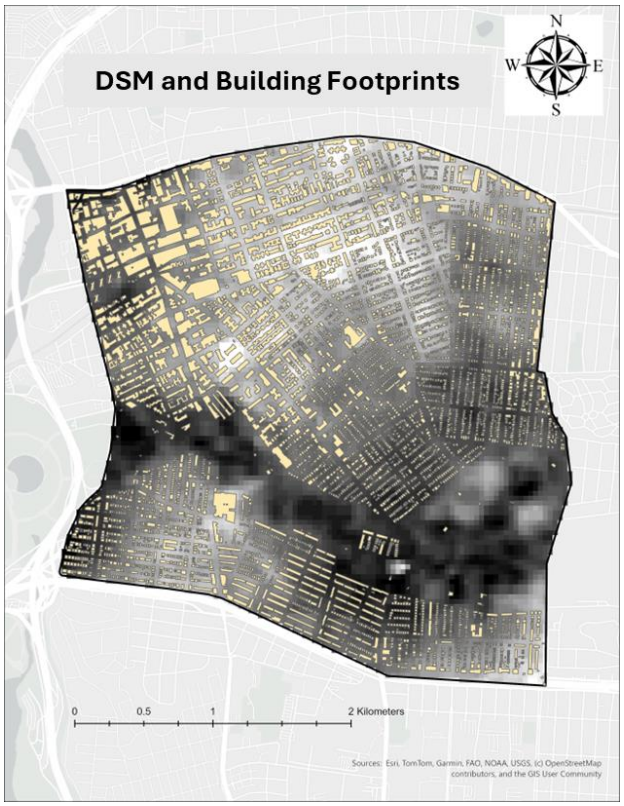

Figure 1: Study area map showing the Kisena Park neighborhood within Queens Borough, New York City. The map highlights the spatial extent of the analysis boundary, surrounding land use context, and key hydrological features relevant to the green roof suitability assessment.

Primary Data Sources

NYC Open Data Portal (data.cityofnewyork.us): Provides building footprints, land use and zoning layers, stormwater infrastructure shapefiles, combined sewer overflow locations, and street centerlines for New York City. All datasets are freely accessible and updated regularly by city agencies including the NYC Department of City Planning and the Department of Environmental Protection.

USGS National Elevation Dataset (NED) / 3DEP: Supplies high-resolution digital elevation model (DEM) data at 1-meter resolution for the study area, enabling topographic analysis, slope computation, and hydrological flow modeling. Accessed through the USGS National Map Downloader at nationalmap.gov.

NASA POWER / NOAA Solar Radiation Data: Provides gridded solar irradiance and photosynthetically active radiation (PAR) data used to assess rooftop solar exposure. The NASA POWER project (power.larc.nasa.gov) offers freely downloadable climate and solar datasets at spatial resolutions suitable for urban-scale analysis.

NYC Department of Buildings (DOB) — Building Information System: Provides construction year, building class, floor area, and permit history for all structures in New York City. This data is used to assess building age and estimate structural feasibility for green roof retrofitting. Accessible at nyc.gov/buildings.

NYC Department of Environmental Protection (DEP) — Green Infrastructure Program: Offers spatial data on existing green infrastructure installations, combined sewer drainage areas, and stormwater retention targets across New York City. This contextual dataset helps situate the Kisena study area within the city's broader green infrastructure strategy and provides benchmarks for expected stormwater retention performance.

OpenStreetMap (OSM) via Overpass API: Supplements official city datasets with rooftop geometry, street network data, and points of interest for cartographic visualization. OSM data is particularly useful for cross-validation of building footprint completeness and for generating basemap layers used in the Tableau dashboard outputs.